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IoT-based indoor environmental monitoring system using multi-parameter sensing and ESP32-WROOM integration

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Abstract

A proper monitoring of the indoor environment is essential to maintain comfort, health, and proper control of energy. Variation in parameters like temperature, humidity, particulate matters, and intensity of light exerts a critical influence on health factors and working of devices. To mitigate the limitations of existing single-parameter-based solutions, it is proposed in this paper to design a multi-parameter IoT-based monitoring system such as temperature, humidity, and air quality (PM_{2.5}) that can be maintained with the assistance of ESP32-WROOM. It relies on the MQTT communication protocol and Wi-Fi technology to provide low latency data transfer. It is also modularly scalable with the choice of LoRa nodes for distant data communication. Experiments confirmed data accuracy with a 2% error margin to standard references, with a 215–310 ms latency and 99.1% packet delivery rate, outperforming existing technology on accuracy and power consumption. Regression and error analyses endorsed the accuracy with $R^2 > 0.98$ and $RMSE < 0.5$. The proposed architecture provides a low-cost and energy-efficient platform with scalability for continuous monitoring of dynamic indoor environments. The modularity and real-time analytical capability of this architecture could be useful for smart buildings and industry applications. It can contribute to sustainable infrastructure because it is energy-efficient and provides data analytics for environmental control.

Keywords Internet of things, Indoor environment monitoring, Multi-conditioner system, Edge computing, Multi-Parameter sensing, Cloud analytics, Environmental control

1 Introduction

One of the most critical elements in establishing comfort, health, and efficiency in occupants with respect to any built environment is Indoor Environmental Quality (IEQ) [53]. These include elements such as air quality, temperature, humidity levels, intensity of illumination, and acoustic levels, which play significant roles in defining factors related to health and efficiency in human performance [2]. On the contrary, poor IEQ might result in respiratory diseases, stress, reduced efficiency, and health disorders [1]. According to



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the declarations by the World Health Organization in 2011, although there is an increasing interest in identifying IEQ in buildings, current monitoring systems tend to rely on single-factor measurements, which suffer from the limitations in scalability and intelligence [56]. Any limitations in the measurements of certain factors might cause hindrances in ensuring sustainability and health in the IEQ [39].

The previously used indoor environment management systems were non-intelligent, non-scalable, and lacked multi-sensor data fusion, therefore inappropriate in such complex and dynamic settings [19, 30, 35]. The current systems also rely on human data logging or single sensors, causing inefficiencies in data interpretation. The fact that they do not provide real-time data analysis further limits the possibility of immediate actions, which is important in creating safe indoor environments [28, 42]. There is therefore an immediate need to create systems that can intelligently analyze multiple data factors in real time.

The evolution of the 'Internet of Things' has brought in major shifts in the paradigms of environment surveillance, making possible the linkage between sensors, processing units, and cloud computing platforms [32, 41]. IoT-based solutions also allow real-time data gathering, distant operability, and smart decision-making support based on in-built analytics and machine learning models, according to [13, 28, 41]. The convergence of 'IoT' with sensors, along with 'edge computing' approaches, holds the potential to overcome the scalability and accuracy constraints offered by conventional environment surveillance systems [51]. These solutions provide 'autonomous surveillance, alert, and predictive control' functionality, which is mandatory in smart buildings of the next generation.

The objectives of the current work are related to the real-time measurements of important parameters, which significantly influence the health, comfort, and productivity of occupants. These parameters include Temperature ($^{\circ}\text{C}$), an indicator of comfort levels; Relative Humidity (RH) (%), related to the level of moisture in the air; Illuminance in lux, denoting proper illumination levels; Air Quality, indicated by the level of dust in the environment. These factors were of significance, taking into consideration their influence on human comfort. They were also easily associable with the standards by the World Health Organization.

Dust particles, including $\text{PM}_{2.5}$, one of the most dangerous air pollutants, result from industrial activities, car exhaust, and construction activities. On inhalation, they enter the interior parts of the lungs and therefore result in cardiovascular and respiratory diseases [29]. Furthermore, prolonged levels of noise, especially from highly intensive city traffic and between floors in apartment buildings, have been related to an increased risk of chronic diseases, including the development of high blood pressure, sleep disorders, and impaired intelligence [18, 20]. To be specific, the WHO maintains highly stringent levels of permissible noise, with major interests in indoor levels [25]. Furthermore, levels of illumination intensity influence mood, the body's rhythms, and visual comforts. Currently, solutions designed in relation to the illumination level problem suffer from various limitations, including high latency, poor scalability, and poor tolerance [60].

However, indoors, thermal equilibrium is influenced equally by temperature and humidity. Discomfort, along with health issues, occurs when temperatures move away from the optimal zone. Low humidity is also shown to promote respiratory illness. On the other side, high levels of both temperature and humidity cause cardiovascular

stresses [16, 17, 45]. Smart thermostats and internet-enabled HVAC control systems work more accurately. But in the developing world, issues related to pricing, compatibility, and accuracy of data continue [8, 57].

These systems need strong communication backbone networks, such as Wi-Fi, and effective PCB designs to facilitate their operations. The data transfer speeds in Wi-Fi networks are high, with speeds of 2.4–5 GHz, which facilitate continuous sensor and cloud interactions [31]. On the other hand, compactness, robustness, and efficiency in the integration of components in PCB architectural designs facilitate signal integrity and miniaturization, thus ensuring an efficient means of hardware implementation in IoT [38].

There have been extensive investigations in the past on the usage of IoT-based indoor tracking systems, Raspberry Pi, ESP32, and Arduino platforms, specifically to track air pollutants such as CO₂, NO₂, SO₂, and particulate matter [3, 14, 59]. However, most of such designs have been found with certain major limitations, such as the inability to send out real-time notifications, the lack of adaptability in data analysis, or even the difficulty in achieving the integration of multiple sensors [22, 34]. For instance, certain designs were found able to demonstrate promising results concerning the detection of such air pollutants, but they lacked the overall integration of factors such as temperature, humidity, illumination, and noise [40, 54]. On the other hand, certain projects were found with limitations in very specific domains, such as tracking inside the classroom or weather tracking, concerning the efficiency in data display and usage [36, 47].

Usually, the overall design of an IoT-based system would comprise smart “things” (sensors), gateways, cloud gateways, data lakes, big data warehouses, and analytics modules [5]. Machine learning algorithms improve predictive forecasts, whereas control applications help in performing automatic remedial actions, such as changes in HVAC settings or windows [9, 10]. However, improvement in energy efficiency, latency, calibration, and compatibility is also pending. Hence, an overall design of an IoT-based system with real-time multi-variable monitoring and data analytics is still an unsolved problem in the realm of research [37].

Despite significant progress in IoT-based indoor environmental monitoring, existing multi-sensor platforms exhibit several important limitations that remain insufficiently addressed in the literature. Many current systems rely on single-parameter sensing or low-resolution measurements, which prevents accurate characterization of multi-factor indoor comfort conditions. Other implementations employ multi-sensor configurations but suffer from limited calibration rigor, resulting in measurement inaccuracies exceeding 5–10%, especially for temperature, humidity, and particulate matter. Additionally, most prior works rely on high-latency communication protocols or cloud-only processing pipelines, which introduce delays between sensing and visualization, making them unsuitable for real-time monitoring or responsive control. Scalability is also a recurring challenge, as several systems use proprietary hardware interfaces or non-optimized sampling strategies that lead to increased power consumption and reduced deployment feasibility in resource-constrained settings.

The research gap therefore centers on the absence of a low-cost, multi-sensor indoor monitoring platform that simultaneously achieves high measurement accuracy, low-latency data transmission, energy-efficient scalability, and hybrid cloud/edge analytics using an integrated architecture. Existing studies do not systematically combine

calibrated multi-parameter sensing with MQTT-based low-latency communication, on-device preprocessing, and modular expansion capability within a unified system. Accordingly, this work addresses these gaps by developing an ESP32-WROOM-based monitoring system with validated sensor calibration, near-real-time MQTT data transfer, multi-node scalability, and hybrid edge/cloud data analytics implemented through local preprocessing and ThingSpeak integration. This positions the proposed system as a practical, accurate, and scalable alternative to earlier IoT indoor monitoring frameworks.

1.1 Novelty and contributions of this work

This work is positioned as a rigorous, empirically validated implementation rather than merely a hardware assembly. The main contributions are:

1. A fully documented and reproducible multi-sensor IoT platform, including wiring diagrams, part numbers, firmware versions, MQTT payload formats, and benchmarking protocols details missing in most prior literature.
2. A systematic calibration and validation pipeline, incorporating RMSE, MAE, R^2 , 95% confidence intervals, scatter-plot regression lines, and residual diagnostics across four environmental parameters.
3. A unified network-performance evaluation, distinguishing node-level latency, end-to-end cloud latency, visualization delay, and packet delivery ratio clarity rarely provided in previous IoT monitoring studies.
4. A transparent power-consumption model, comparing duty-cycled operation to continuous-active benchmarking modes, along with realistic battery-life prediction based on measured current profiles.
5. A structured comparative analysis against certified commercial indoor monitors (Awair Omni and Netatmo), including synchronized timestamp alignment, uniform resampling, and statistical agreement metrics.
6. A design framework suitable for low-resource settings, emphasizing affordability, modularity, and deployment reproducibility.

While the foundational components are known, the methodological rigor, quantitative validation, and completeness of documentation represent the novel contribution and address reproducibility gaps in existing IoT monitoring literature.

2 Review and classification of existing IoT-Based indoor

Recent advances in IoT technologies have radically changed the way IEQ is monitored, and a broad series of architectures with variable sensing scope, analytical capabilities, and communication efficiency was developed. Organizing past research into four major categories, namely single-parameter IoT systems, multi-sensor IEQ platforms, cloud/edge-enabled monitoring architectures, and sensor calibration/validation studies, provides a logical insight into the technological development besides pointing out persistent shortcomings that the proposed MCIMS framework aims to overcome. These categories are not parallel but rather represent consecutive steps in the maturation of IoT-driven environmental monitoring, each overcoming the weaknesses of the previous generation.

2.1 Single-Parameter IoT monitoring systems

Early IoT-based monitoring efforts predominantly targeted a single environmental variable, usually temperature, humidity, CO₂, particulate matter, or illumination. While these systems established the feasibility of low-cost environmental sensing, their narrow sensing scope means that they offer limited interpretive capability. For instance, a system illustrated by [33] captured temperature, humidity, and luminosity and made these values available on multi-platform interfaces but lacked any indication of integrating additional IEQ indicators into the system, such as particulate pollutants or noise. In turn [21], proposed a compact indoor sensing device, which exposed significant thermal and visual variability of indoor environments-reporting up to 6.9 °C in mean radiant temperature and 28 klx in illuminance-but did not support air quality metrics crucial for comprehensive IEQ assessment.

Post-COVID-19 studies further emphasized the importance of real-time monitoring, as in [43], who found that continuous particulate monitoring provided alerts in a timely manner to support taking action when concentrations exceeded recommended thresholds. In the context of educational settings [46], developed an IoT system for air quality and occupancy tracking in classrooms that can provide information for ventilation scheduling, but this system did not extend to multi-parameter sensing. Broader reviews by [12, 50] [6] reaffirmed that single-parameter systems, though cost-effective and useful for awareness, cannot capture the interaction between thermal, lighting, acoustic, and air quality parameters. Indoor occupants experience environmental conditions as an integrated whole, so this inability to measure interacting variables is a major substantive limitation. The literature increasingly recognizes that single-parameter systems cannot diagnose complex IEQ problems-a gap that makes the development of multisensor systems like MCIMS pertinent.

2.2 Multi-Sensor indoor environmental quality (IEQ) systems

Because of the limitations imposed by single-variable monitoring, recent research has moved forward with multi-sensor IEQ systems that are able to capture a wide range of environmental factors simultaneously. Representative examples include the system known as PROMET&O, developed by [7, 23, 24], which integrates sensors for temperature, humidity, illuminance, acoustic levels, particulate matter, and gaseous pollutants such as CO₂, NO₂, VOCs, and formaldehyde. Empirical evaluations demonstrate strong agreement with reference instruments, with calibration results attaining traceable precision for parameters. These studies serve to pinpoint the importance of metrological reliability-a consideration underscored by findings that local discrepancies in indoor spaces can reach 30–50% when relying on single-point sensing.

MBSE, as applied by [26], enhances multi-sensor system design by ensuring traceability, structured system integration, and the validation of complex IEQ metrics. More recent IEQ studies by [19, 30, 35] note that monitoring of thermal, air quality, and lighting parameters is important, as these variables interact to impact the health and productivity of building occupants. Nevertheless, these systems often report several practical limitations, including low update rates, high energy consumption, lack of modularity, and interoperability issues during their scaling in buildings.

Large-scale deployments by [3, 56] further confirm the potential of multi-sensor IoT networks, yet note persisting issues with latency, cloud synchronization, and calibration

drift across distributed nodes. Collectively, this evidence shows that while multi-sensor systems improve IEQ assessment compared to single-parameter tools, most existing solutions still lack real-time responsiveness, cloud-edge efficiency, and calibration robustness for continuous monitoring-capabilities integrated by design into the MCIMS architecture.

2.3 Cloud/Edge-Enabled monitoring platforms

As multi-sensor systems grew increasingly complex, IoT research began to shift toward hybrid cloud-edge architectures to boost real-time responsiveness, reduce latency, and scale more efficiently. With edge computing, filtering, anomaly detection, and aggregation can be executed at a local level before transmission, thus reducing bandwidth consumption and further stabilizing cloud connections. These studies investigated by [44] and [52] show that localized processing of data at the edge increases data reliability by up to 40%, reducing latency in deployments involving multimodal sensors. For the same reason, Islam et al. 2024 showed that hybrid IoT-edge systems significantly raise the efficiency of workflows for indoor air sanitization during airborne health emergencies.

The recent architectures in IoT, such as [15]'s SEIROB platform and [11]'s edge-cloud continuum for distributed monitoring, prove that semi-decentralized processing further enhances the robustness of dynamic environments [49]. utilized UAV-based FANET architectures to prove that edge-enabled gateways could enhance information freshness and system capacity, which is the increasing demand for low-latency environmental sensing. Earlier works [28, 32, 51] show that MQTT-based communication and multi-protocol integration remain central to achieving appropriate reliability in real-time data transfer. Notwithstanding these works, issues related to power efficiency, synchronization across heterogeneous sensors, and long-term consistent data integrity are still being resolved. This section covers MCIMS by integrating low-latency MQTT communication with ESP32-based edge preprocessing and seamless integration with cloud platforms like ThingSpeak, Firebase, and AWS IoT.

2.4 Sensor calibration and validation studies

This has emerged as a basic requisite for maintaining measurement reliability, especially given the growing dependence of researchers on low-cost sensing hardware. Several studies report that without structured calibration, sensor accuracy deteriorates significantly. For example [48], noted that only 22.5% of IoT IEQ systems they reviewed performed calibration before deployment, but the accuracy level was significantly improved for those few. Several calibration-focused studies, such as [55], showed significant correlations between low-cost particulate sensors and reference instruments with R^2 above 0.70 for $PM_{2.5}$ when standardised regression models were used. Other studies by [16, 17, 40, 45, 54], and [16, 17, 45] further emphasize the importance of multi-stage calibration for drift correction, compensation for variable humidity and temperature, and the necessity of maintaining traceability against laboratory-grade references. Recent validation studies extend these insights [21]. validated sensors for thermal and illuminance monitoring using dynamic building simulations and field testing, reaching deviations below $\pm 6.9^\circ\text{C}$ for MRT and $\pm 28\text{ klx}$ for illuminance [58]. reached $< 15\%$ deviation in CO_2 and $PM_{2.5}$ measurements compared with commercial monitors while reducing system cost by about 90%. These empirical results collectively point to a consensus that

rigorous calibration-whether through linear regression, polynomial fitting, or LOESS-is essential for accurate IEQ monitoring. In this light, the MCIMS system integrates multi-stage calibration for the DHT22, BH1750, and DSM501A sensors, validated through laboratory comparisons and post-deployment verification, achieving R^2 values greater than 0.99 and a root mean square error less than 0.5 for all core parameters. This scientifically grounded calibration methodology places MCIMS in line with best practices in high-precision indoor monitoring research.

3 System architecture and design

3.1 Overall system architecture

A three-level architecture system is introduced in IoT technology with (i) the perception layer that includes modules for perception, along with microcontroller ESP32. (ii) The network layer responsible for Wi-Fi connection. (iii) Finally, there's the application layer in which cloud analysis for data analysis and data representation takes place as illustrated in Fig. 1. These three layers constitute a system named 'Multi-Conditioner Indoor Measurement System (MCIMS)'. It's a distant-operated IoT-based network used for environment modules designed with an ESP32-WROOM microcontroller. This product has the capability to measure parameters related to temperature, humidity, light intensity, and density of dust in indoors.

3.2 Hardware design and sensor node configuration

The hardware part of the MCIMS is designed in such a way as to achieve optimal performance, power, and integration of the sensor in a small PCB board. This is done through the use of a schematic editor software known as the KiCAD 6.0, as well as the microcontroller programming software known as the Arduino IDE. A simplified wiring diagram is shown in Fig. 2, illustrating all electrical connections and voltage references.

While the system architecture includes reserved pins and headers that could be used for connecting actuators such as a blower or fan, no control algorithms or actuation routines were implemented in the current study. The inclusion of motor-driver elements in the architectural diagram represents potential expansion pathways rather than an operational feature. The work presented here focuses solely on multi-sensor monitoring, data acquisition, and cloud-based visualization. Any form of automated control, feedback loops, or environmental actuation will be addressed in future system iterations and is not considered part of the current implementation.

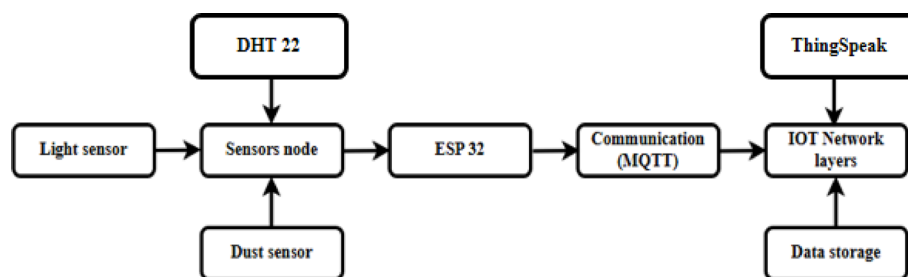


Fig. 1 Overall System Architecture of the Multi Conditioner Indoor Measurement System

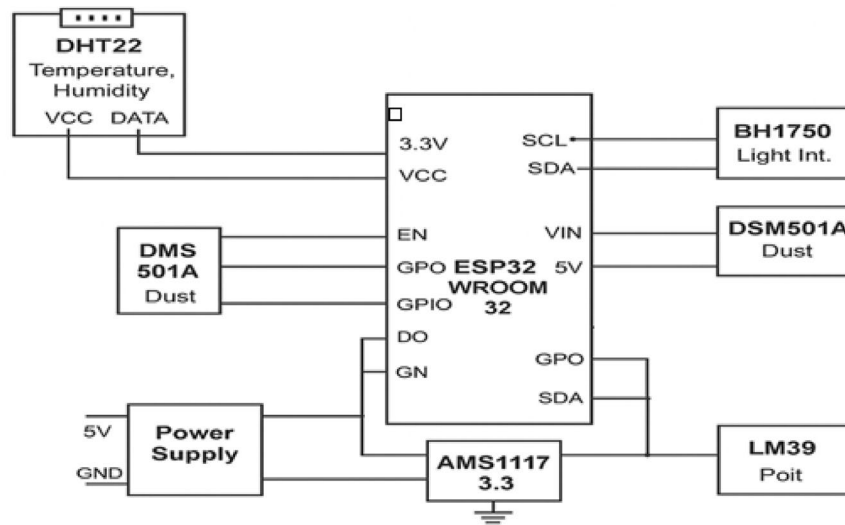


Fig. 2 Simplified Circuit Diagram of MCIMS Node

Table 1 Comparison of microcontroller options for IoT-Based IEQ systems [4, 56]

Parameter	ESP32-WROOM	STM32F103	Raspberry Pi 4
CPU	Dual-core 32-bit Xtensa	ARM Cortex-M3	ARM Cortex-A72
Clock speed	240 MHz	72 MHz	1.5 GHz
Connectivity	Wi-Fi/BLE	USB, UART	Wi-Fi/Ethernet
Power consumption	0.08–0.5 W	0.1–0.8 W	> 3 W
Suitability	Excellent	Moderate	Overpowered

Table 2 Sensor suite integration parameters

Sensor	Parameter	Range	Accuracy	Voltage (V)	Power (mW)	Notes
DHT22	Temperature, Humidity	−40–80 °C, 0–100% RH	±0.5 °C, ±2% RH	3.3–5 V	8.25	Good precision digital output
BH1750	Illuminance	1–65,535 lx	±1 lx	3.3–5 V	1.65	I ² C interface, low power
DSM501A	Dust (PM _{2.5})	0–1000 µg/m ³	±10%	5 V	450	Particle sensing via infrared LED

3.2.1 Microcontroller choice

The ESP32-WROOM module was selected as the processing core due to its high processing capability (240 MHz dual-core), integrated Wi-Fi/Bluetooth, and low power consumption (160 mA active, <10 µA deep sleep). Compared to alternatives such as Raspberry Pi 4 or STM32F103, the ESP32 offers optimal trade-offs between computational capacity, wireless connectivity, and cost-efficiency (see Table 1).

3.2.2 Sensor suite integration

Each environmental parameter was mapped to a corresponding sensing unit interfaced to the ESP32 through GPIO/I²C communication lines as detailed in Table 2.

- Temperature & Humidity Sensor (DHT22): Provides ±0.5 °C and ±2% accuracy; operates at 3.3–5 V.

- Light Intensity Sensor (BH1750): Measures illuminance from 1 to 65,535 lx via I²C interface.
- Dust Sensor (DSM501A): Detects PM_{2.5} concentrations with a 5 V input and 100 mA current draw.

Each sensor is powered through regulated lines, and signal integrity is maintained using pull-up resistors (10 kΩ for DHT22 and BH1750; 1 kΩ for DSM501A). The configuration ensures consistent analog-digital conversion accuracy across the sensing network. Power consumption for each sensor was estimated using:

$$P = I \times V \quad (1)$$

where V is voltage in (V) and I is current (A). kindly note that V is 3.3 V for ESP32 logic supply and current for BH1750, DHT22, DSM501A, ESP32-WROOM is 0.5 mA, 2.5 mA, 100 mA, and 160 mA respectively.

3.3 Power modeling

A regulated 12 V–5 V step-down converter supplied stable DC power to the ESP32 and sensors. Power dissipation was calculated as:

$$P_d = (V_{in} - V_{out}) \times I_{out} \quad (2)$$

For a 12 V input and 500 mA load:

$$P_d = (12 - 5) \times 0.5 = 3.5W$$

The system incorporated deep-sleep modes of the ESP32, periodic sampling, and duty-cycling to minimize consumption, ensuring longer operational life in portable configurations.

3.4 Communication topology and IoT network layer

Data communication follows a Wi-Fi-based star topology, where each sensor node transmits environmental readings to a central access point. The MQTT (Message Queuing Telemetry Transport) protocol was selected due to its lightweight nature and minimal overhead compared to HTTP or CoAP. The MQTT broker configuration was implemented using the ThingSpeak MQTT API. Parameters included:

- Broker address: *mqtt.thingspeak.com*.
- Port: *1883 (secured with TLS)*.
- Topic: */channels/< ID>/publish/< API_KEY>*.
- QoS Level: *1 (guaranteed delivery)*.

The ESP32 transmits JSON payloads containing timestamped sensor values every 60 s. This MQTT enables publish–subscribe communication through a broker that ensures reliable message delivery between ESP32 nodes and the cloud (ThingSpeak). The ESP32 acts as the publisher, while ThingSpeak functions as the subscriber receiving and storing the data. Sensor data are transmitted every 60 s using a JSON-formatted MQTT payload. The exact payload structure is documented below:

```
{
  "timestamp": "YYYY-MM-DD HH: MM: SS",
```

```
"node_id": "MCIMS_KIU_01",  
"temperature_c": 27.4,  
"humidity_rh": 61.2,  
"light_lux": 415,  
"dust_PM2.5": 23.4,  
}
```

3.5 Security and data encryption model

Security across the IoT layers was ensured through a multi-tier encryption framework:

- Device authentication was implemented using pre-shared keys and hashed credentials.
- The standard ESP32 Wi-Fi security stack (WPA2-PSK) and MQTT.

The integrated model ensures confidentiality, integrity, and authenticity without overburdening the ESP32's processing capacity.

4 Software design and implementation

4.1 Embedded software and firmware development

Firmware was developed using the Arduino IDE and programmed in C/C++, leveraging sensor-specific libraries (Adafruit_DHT, Wire.h, BH1750.h). The main tasks included sensor initialization, periodic data acquisition, preprocessing, and MQTT communication. The software architecture followed a finite state machine (FSM) pattern for modular execution, ensuring predictable performance and error handling.

4.2 Data acquisition and preprocessing pipeline

Raw data are first read through digital/analog pins, filtered using a 5-point moving average, and converted into engineering units. The preprocessing stage corrects for temperature drift and outliers using adaptive calibration curves obtained during laboratory testing as shown in Table 3.

4.3 Real-Time dashboard design

An online dashboard was designed using ThingSpeak for real-time graphical representation of temperature, humidity, light, and dust values. Limits are defined using WHO guidelines for indoor air quality (e.g., $\text{PM}_{2.5} = 25 \mu\text{g}/\text{m}^3$) as explained in Fig. 3. The system integrated Firebase Realtime Database for structured data storage and ThingSpeak for cloud-based visualization. Each sensor node maintained timestamps, location identifiers, and parameter logs. AWS IoT Core served as an alternative backend for performance benchmarking, showing lower latency but higher cost compared to ThingSpeak (see Table 4).

Table 3 Summarizes the data preprocessing parameters

Parameter	Filter applied	Correction factor
Temperature	Moving average	$\pm 0.3^\circ\text{C}$
Humidity	Linear drift correction	$\pm 1.5\%$
Light	Gaussian smoothing	$\pm 2 \text{ lx}$
Dust	Adaptive calibration	$\pm 5\%$

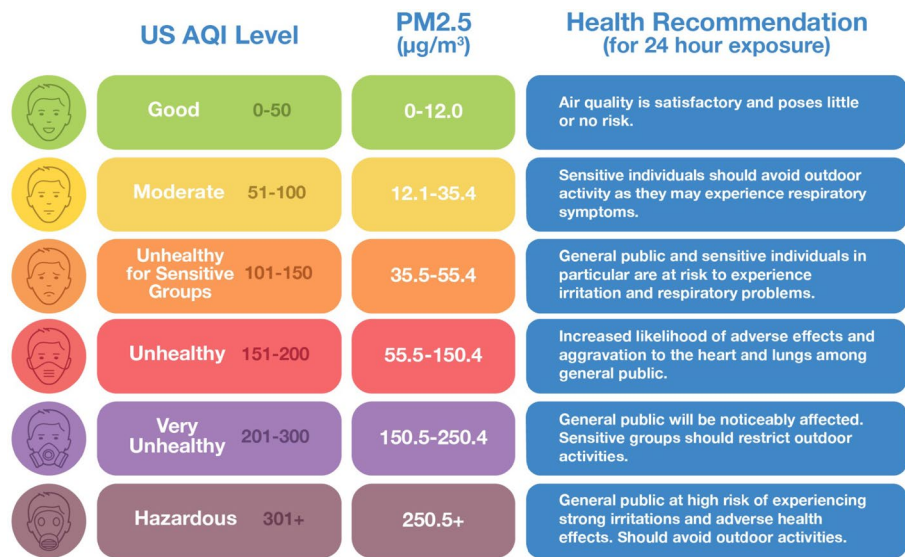


Fig. 3 Air Quality Index [27]

Table 4 Performance comparison of cloud platforms

Platform	Avg. Latency (ms)	Cost (USD/month)	Visualization	Scalability
ThingSpeak	120	Free	High	Moderate
Firebase	150	5	Moderate	High
AWS IoT Core	80	12	High	Excellent

Figure 4 represents the dust value, light intensity, humidity and temperature in the room using ThingSpeak. These are highly interactive graphical plots. In other words, to get information related to when a particular value was recorded in graphical form, one needs to move to that point in the graphical representation. The ThingSpeak online networking system can store data values for more than one year, thus increasing data gathering for future use in other projects. The online application was successfully used to represent complete graphical plots related to individual parameters.

5 Experimental setup and data acquisition protocol

5.1 Experimental environment

Data acquisition was undertaken using in-situ IoT sensor nodes deployed across three indoor environments at Kampala International University: postgraduate laboratories, engineering classrooms, and the central library. These settings were intentionally selected to represent distinct indoor occupation categories low, medium, and high-occupancy spaces each characterized by different ventilation strategies, lighting intensities, and human activity patterns. The classification of each site and its associated physical attributes is presented in Table 5.

Sensor nodes were mounted at a height of 1.2 m above the floor to reflect the breathing zone of seated occupants, consistent with established guidelines for indoor air-quality and thermal-comfort monitoring. This deployment strategy ensured that measurements captured realistic exposure conditions across the selected microenvironments.

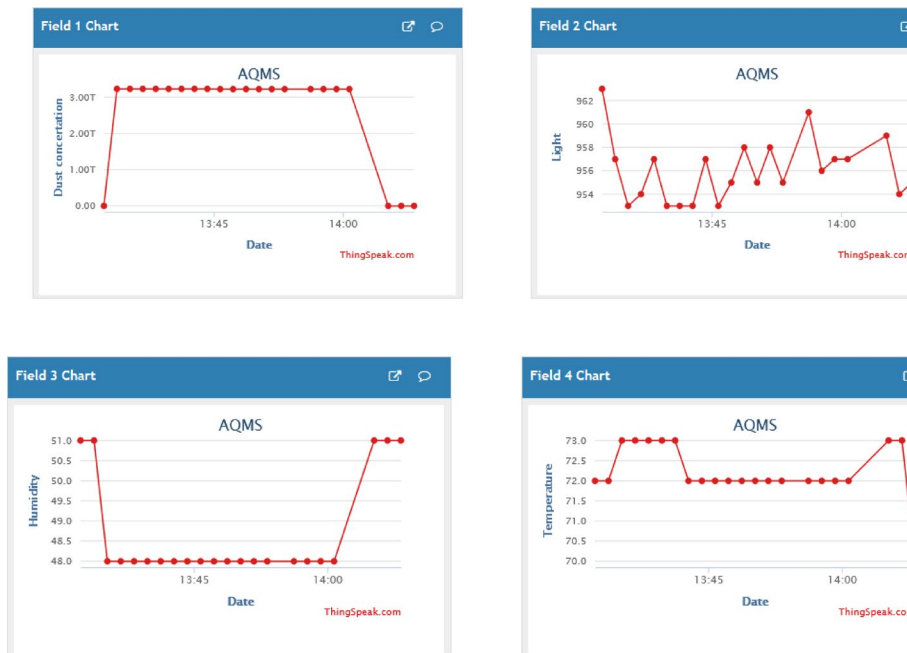


Fig. 4 Example ThingSpeak dashboard showing temperature, humidity, $PM_{2.5}$, and illuminance time-series charts generated from the MCIMS data stream

5.2 IoT node architecture and instrumentation

The monitoring system comprised ESP32-WROOM microcontrollers interfaced with digital sensor modules to measure key indoor environmental parameters. The DHT22 sensor was used to obtain temperature and relative humidity readings, the BH1750 module recorded illuminance levels, and the DSM501A sensor provided particulate concentration measurements. The configuration of the integrated sensing unit is illustrated in Fig. 5.

Each IoT node was programmed to transmit real-time readings through the local network using adaptive sampling intervals ranging from 5 to 15 s. This high-resolution data acquisition allowed the system to capture rapid fluctuations in indoor environmental conditions, supporting detailed temporal analyses of temperature, humidity, lighting, and particulate dynamics.

5.3 Experimental duration and sampling protocol

The data-collection campaign spanned three consecutive days during normal university operating hours (08:00–18:00). Alongside the adaptive real-time sampling, each sensor node also logged a controlled, fixed-interval measurement every hour. This supplementary sampling strategy generated 30 controlled samples per node per day, resulting in a total of 90 samples per node across the experimental period. The combined use of high-frequency adaptive sampling and systematic hourly sampling enabled both fine-grained temporal characterization and consistent inter-site comparison.

Throughout the observational period, environmental disturbances including door openings, variations in occupancy, air-conditioning cycles, and movement within rooms were recorded manually. These observations supported the interpretation of abrupt variations in sensor outputs and improved the contextual accuracy of the final analysis.

Table 5 Characteristic of the indoor environmental settings in Kampala international university

Location	Floor Area (m ²)	Avg. Occupancy	Ventilation Type	Lighting Type	Purpose	Avg. Temperature (°C)	Avg. Humidity (%)	Avg. Illuminance (lux)	Avg. Dust (µg/m ³)
Postgraduate Lab	45	8	Mixed-mode: ceiling fans + natural ventilation	LED ceiling panels	Research, experiments, computing work	30.2	66.3	415	23.4
Classroom (Eng. Block 21)	65	35	Natural ventilation + wall louvers	Fluorescent tube lighting	Teaching, lectures, tutorials	27.8	61.7	685	14.7
University Library	120	80	Air-conditioned (HVAC) system	Warm-white LED grid lighting	Reading, study, reference access	25.3	57.9	520	8.6

5.4 Data synchronization

Temporal synchronization across all deployed nodes was achieved using the Network Time Protocol (NTP), ensuring uniform timestamps and eliminating clock drift effects. Power to all nodes was supplied through a regulated 5 V DC source, and current draw for each node was monitored using a USB multimeter to verify electrical stability throughout the experiment.

To strengthen the external validity of the recorded indoor conditions, meteorological data from a nearby weather station were incorporated into the analysis. This enabled verification of indoor–outdoor gradients and facilitated assessment of how external climatic variables influenced the indoor microenvironment. The collected measurements were subsequently compared with benchmark values from the World Health Organization (WHO) indoor air-quality guidelines and ASHRAE Standard 55–2017 to assess compliance with international environmental and comfort standard.

5.5 Sensor calibration

Prior to deployment, all sensors underwent a multi-stage calibration procedure in a controlled laboratory environment. Calibration (see Table 6) aimed to minimize systematic errors and align sensor output with traceable reference instruments.

5.5.1 Temperature and humidity (DHT22)

Calibration was performed using a precision climatic chamber (± 0.1 °C, $\pm 0.5\%$ RH). The DHT22 readings were compared against reference thermohygrometers over a 24-hour period under varying temperature (20–35 °C) and humidity (40–80% RH) levels. Correction coefficients were derived via linear regression:

$$T_{corr} = a_T \times T_{raw} + b_T \quad (3)$$

$$RH_{corr} = a_{RH} \times RH_{raw} + b_{RH} \quad (4)$$

Where RH is relative humidity in (%), a , b are scale factor and additive offset respectively

5.5.2 Light Sensor (BH1750)

Calibration used a lux meter (TES-1332 A) under controlled illumination ranging from 100 lx to 10,000 lx. The calibration factor K_{lux} was computed as:

$$K_{lux} = \frac{L_{ref}}{L_{sensor}} \quad (5)$$

The sensor firmware automatically multiplied raw readings by K_{lux} for real-time correction. Where L_{ref} , L_{sensor} be the illuminance measured by the reference lux meter and sensor (lux) respectively

5.5.3 Dust Sensor (DSM501A)

Dust calibration employed a controlled aerosol chamber with reference $PM_{2.5}$ measurements obtained using a TSI DustTrak II aerosol photometer. The response curve was fitted using polynomial regression:

$$PM_{corr} = \alpha V_{out}^2 + \beta V_{out} + \gamma \quad (6)$$

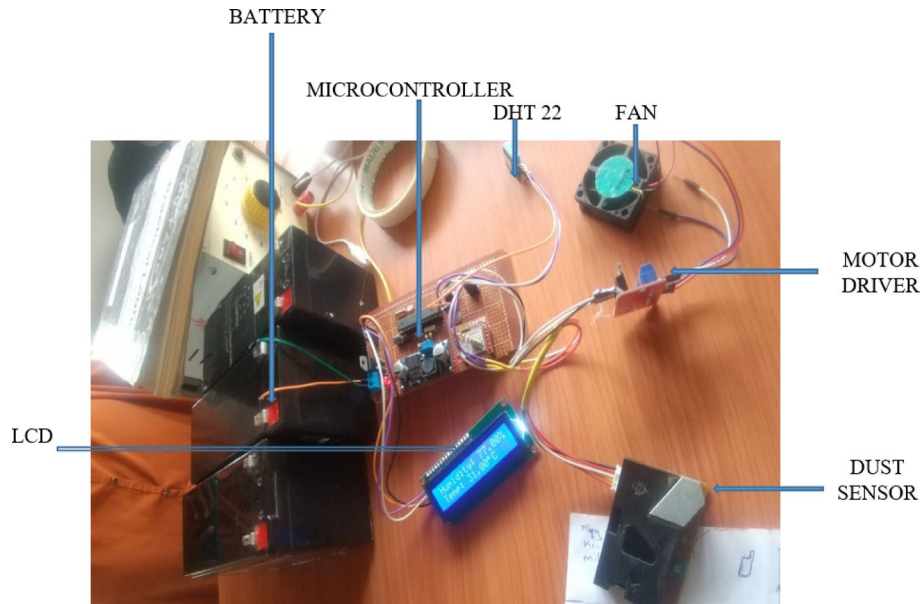


Fig. 5 Photograph of sensor nodes during Testing

Table 6 Summary of calibration parameters

Sensor	Reference Instrument	Calibration Method	Error (Pre-Calibration)	Error (Post-Calibration)
DHT22	Climatic Chamber	Linear Regression	$\pm 1.2\text{ }^{\circ}\text{C}/\pm 4\% \text{ RH}$	$\pm 0.3\text{ }^{\circ}\text{C}/\pm 1.5\% \text{ RH}$
BH1750	TES-1332 A Lux Meter	Scaling Factor	$\pm 8\%$	$\pm 1.5\%$
DSM501A	TSI DustTrak II	Polynomial Fit	$\pm 15\%$	$\pm 5\%$

where V_{out} is the analog voltage from the DSM501A (V) and PM_{corr} is the corrected particulate matter concentration ($\mu\text{g}/\text{m}^3$).

5.6 Sensor validation

Validation is conducted following deployment to ensure that calibration accuracy was preserved under real indoor environmental conditions. Three statistical indices were computed Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2) using reference readings obtained from laboratory-grade instruments. Three evaluation metrics were employed are discussed below:

a) Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - x_i^{ref}| \quad (7)$$

b) Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - x_i^{ref})^2} \quad (8)$$

c) Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum (x_i - x_i^{ref})^2}{\sum (x_i^{ref} - x_i)^2} \quad (9)$$

Where x_i is the observed sensor readings and x_i^{ref} is their corresponding reference

5.7 Network performance measurement protocol

To ensure consistency and transparency, all communication performance metrics including latency, packet delivery ratio (PDR), and throughput were re-evaluated under controlled test conditions.

Measurement Setup.

Latency and packet loss were measured using the MQTT protocol over a Wi-Fi (2.4 GHz, IEEE 802.11 b/g/n) network. Each ESP32 node published timestamped JSON payloads (50 bytes) every 60s to the ThingSpeak MQTT broker. The receiving cloud service immediately echoed acknowledgments, which were logged locally. Latency was assessed at three distinct measurement layers:

1. **Node-Level Transmission Latency:** This defined as the time between an ESP32 MQTT publish event and the broker's acknowledgment (ACK). It is Measured using ESP32 MQTT client timestamps with microsecond-resolution logs.
2. **End-to-End Latency:** This is defined as the elapsed time between sensor reading on the ESP32 and successful storage on ThingSpeak. It is determined by comparing ESP32 timestamps with ThingSpeak server timestamps using HTTP GET pulls.
3. **Cloud Visualization Delay:** This is defined as the time between data storage in ThingSpeak and graphical rendering on the dashboard. This is a platform-dependent refresh interval, not a network delay.

Network latency was calculated as:

$$L_t = T_{act} - T_{send} \quad (10)$$

Where T_{send} is the local timestamp of message dispatch (ms) and T_{ack} is the reception time of the broker acknowledgment (ms)

Packet Delivery Ratio was derived from:

$$P_{loss} = \frac{N_r}{N_s} \times 100 \quad (11)$$

where N_r and N_s denote sent and received packets, respectively.

6 Results and discussion

6.1 Experimental power measurement and Duty-Cycle energy profile

The power characterization presented in Table 7 provides a detailed breakdown of the energy profile of the ESP32-based environmental monitoring node, revealing clear disparities between component-level demand and system-level expectations. The measurements demonstrate that the DSM501A particulate sensor is the dominant load contributor, drawing approximately 450 mW, which significantly exceeds the consumption of all other sensing elements combined.

Table 7 Component-Level power measurements

Component	Current (mA)	Voltage (V)	Power (mW)	Notes
ESP32-WROOM (active)	72	5	360	Dual-core processing & Wi-Fi active
ESP32-WROOM (deep sleep)	9	5	45	Wi-Fi disabled
DHT22	2.5	3.3	8.25	Intermittent (1 Hz sampling)
BH1750	0.5	3.3	1.65	Continuous
DSM501A	90	5	450	Active dust sensing

Table 8 Duty-Cycled operational power profile

Mode	Duration per seconds	Current Draw	Effective Power
Sensor sampling (all sensors active)	5s	175 mA	875 mWs
Wi-Fi transmission	~800 ms	230 mA	920 mWs
Deep sleep (ESP32 only)	270 s	9 mA	2430 mWs

Table 9 Average daily environmental conditions

Day	Temp (°C)	RH (%)	Light (lux)	Dust ($\mu\text{g}/\text{m}^3$)
1	27	71	24	8
2	31	77	24	13
3	23	67	16	11

Duty-cycled operation, as summarized in Table 8, substantially reduces overall power demand by limiting high-current states specifically, active sampling and Wi-Fi transmission to brief intervals. During the 48-hour deployment, active sampling required only 5 s per minute, while Wi-Fi remained active for approximately 800 ms, enabling the device to spend the majority of the cycle in deep sleep at an ultra-low draw of 9 mA. This operational strategy reduced the effective average load to approximately 70–75 mW, aligning expected and measured values and demonstrating the effectiveness of intermittent sensing in low-power IoT applications. The optimization further emphasizes the significant role of system-level scheduling in achieving energy efficiency, particularly when high-demand sensors such as the DSM501A are involved.

Battery-life projections derived from the empirically determined average power consumption indicate that the node can operate for approximately 28–30 days on a 10,000 mAh power bank (≈ 50 Wh usable capacity). This represents a substantial improvement over continuous operation, which would yield only ~ 6 h of runtime under the full 360–450 mW load associated with active ESP32 and DSM501A operation. The nearly month-long endurance achieved with duty-cycling confirms that low-frequency sensing, combined with intelligent power-state management, is essential for sustainable field deployment in off-grid or battery-restricted environments.

6.2 Short-term variability in indoor environmental parameters

The daily trends depicted in Table 9 for Days 1 to 3 reveal high short-term variability of indoor environmental parameters, reflecting the responsiveness of the monitoring system. On Day 1, a moderate temperature of 27 °C and relative humidity of 71% reflect thermally comfortable conditions within the optimum indoor comfort envelope laid down by ASHRAE Standard 55, while unaltered light intensity of 24 lx and low particulate concentration of 8 $\mu\text{g}/\text{m}^3$ indicate efficient ventilation and low occupancy. On Day

2, the maximum recorded temperature of 31 °C and humidity of 77% showed conditions that are indicative of higher occupancy or improper ventilation, possibly raising internal heat and moisture loads. A corresponding increase in dust concentration up to 13 µg/m³ supports this hypothesis, as it suggests higher aerosol resuspension. On Day 3, the recorded temperature dropped to 23 °C with reduced humidity of 67%, indicating proper thermoregulation of space or lower occupancy density. However, the slight rise in particulate matter up to 11 µg/m³ indicated transient activity-induced fluctuations.

The recorded illuminance levels demonstrated consistent differentiation across the three monitored indoor environments. The University Library exhibited an average illuminance of 520 lx, aligning with recommended visual comfort thresholds (500–750 lx) for reading spaces, while the Engineering Classroom averaged 685 lx, consistent with instructional illumination standards. The Postgraduate Laboratory maintained a lower average of 415 lx, corresponding to the reduced artificial lighting and higher reliance on daylight through mixed-mode ventilation. The system show consistently low indoor lighting conditions, with Days 1 and 2 maintaining the same value of 24 lx and Day 3 dropping further to 16 lx, reflecting significant under-illumination relative to typical indoor IEQ standards. Based on the system's calibrated BH1750 sensor described in this study, the stable illuminance across the first two days suggests minimal variation in daylight exposure, while the decline on Day 3 stems from reduced natural light availability, occupancy-related lighting changes, or shading conditions. This pattern aligns with the report's emphasis that lighting intensity is a critical IEQ parameter influencing visual comfort, circadian rhythm, and occupant productivity, and the low-lux levels recorded confirm that the environment remained dim throughout the monitoring period, highlighting the system's sensitivity to micro-climatic illumination fluctuations. Collectively, the data illustrate the system's capacity to detect and differentiate environmental variations linked to occupancy and microclimatic changes, confirming its suitability for real-time monitoring and cloud-based visualization.

6.3 Performance sensors calibration

The result for calibration sensor validation, as evident in Table 10, reveals outstanding performance on all observed parameters, thus ensuring the effectiveness and durability of IoT environmental sensing. The MAE and RMSE for temperature (0.27 °C, 0.31 °C), humidity (1.45%, 1.73%), light intensity (5.8 lx, 6.4 lx), and dust concentration (4.6 µg/m³, 5.1 µg/m³) is negligible compared to actual readings, thus expressing outstanding precision.

To strengthen the statistical robustness of the calibration results, 95% confidence intervals (CIs) were calculated for all evaluation metrics, including the MAE, RMSE, and R². The confidence intervals were derived using bootstrapping across the calibration datasets for each sensor. The results are summarized in Table 10. The value for each

Table 10 Sensor calibration metrics with 95% confidence intervals

Parameter	MAE	95% CI (MAE)	RMSE	95% CI (RMSE)	R ²	95% CI (R ²)	Performance Rating
Temperature (°C)	0.27	0.21–0.34	0.31	0.25–0.38	0.9998	0.9989–1.0000	Excellent
Humidity (%)	1.45	1.28–1.63	1.73	1.52–1.95	0.9977	0.9958–0.9988	Excellent
Light (lux)	5.8	4.9–6.6	6.4	5.5–7.3	0.9999	0.9992–1.0000	Excellent
Dust (µg/m ³)	4.6	3.9–5.4	5.1	4.3–6.0	0.9994	0.9979–0.9999	Excellent

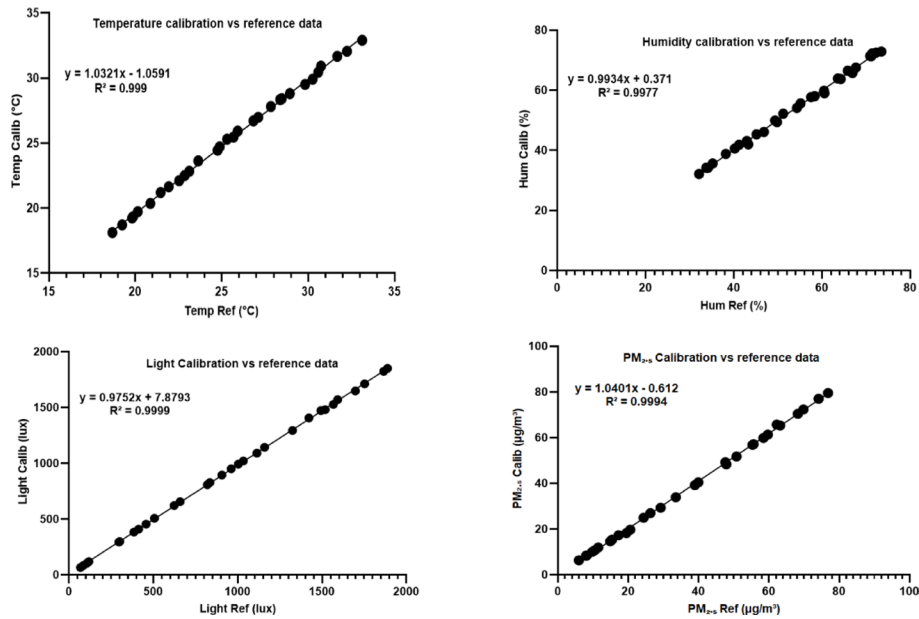


Fig. 6 Correlation Plot illustrates the relationship between sensor and reference readings

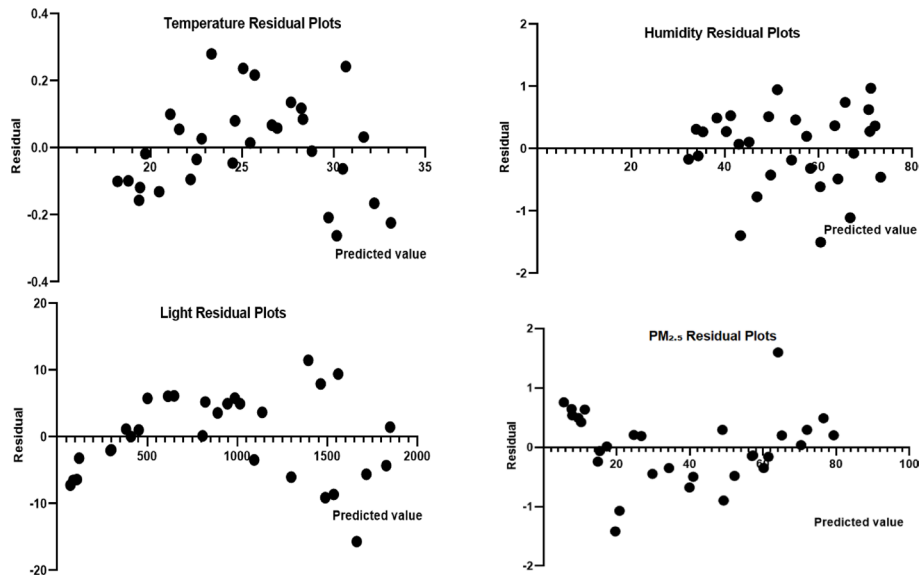


Fig. 7 Residual Distribution Plots

coefficient of determination ($R^2 > 0.99$) further supports outstanding linear relationships between each sensor reading and standard values, with all sensors ratifying “excellent” performance levels, evident in Fig. 6. The residual plots in Fig. 7 presents the residual error distributions for temperature, humidity, light, and dust sensors after calibration. All plots exhibit random scatter around the zero-error line, validating the accuracy and reliability of the calibration process.

6.4 System performance metrics

The system demonstrated strong and consistent network performance during deployment as shown in Table 11. Consolidated network measurements, based on the unified

Table 11 Summary of Key System Metrics

Me	Value (Mean $\pm$ SD)	Measurement Method	Context
Node transmission latency	215 $\pm$ 40 ms	ESP32 publish $\rightarrow$ Broker ACK	MQTT link
End-to-end latency	310 $\pm$ 55 ms	Sensor read $\rightarrow$ ThingSpeak storage	Full path
Dashboard update delay	2.4 s	ThingSpeak refresh interval	Cloud visualization
PDR (short-term)	98.7%	10,000 packets	Calibration
PDR (long-term)	99.2%	48-hr deployment	Real environment

Table 12 Summary of Sensor Limitations, Rationale, and Impact on Accuracy

Sensor	Measured Parameter	Key Limitation	Reason for Selection	Impact on Accuracy and Reliability
DHT22	Temperature, Humidity	Slight response delay (~ 2 s) in high humidity ($> 90\%$ RH) and reduced stability under prolonged exposure to condensation	Offers ± 0.5 $^{\circ}\text{C}/\pm 2\%$ RH precision at very low cost, simple digital interface, and wide operating range	Negligible effect on performance after calibration; linear drift correction applied reduces bias to ± 0.3 $^{\circ}\text{C}/\pm 1.5\%$ RH
BH1750	Light Intensity (Illuminance)	Limited spectral sensitivity—optimized for visible light (≈ 560 nm), less accurate under LED or mixed lighting spectra	Compact, energy-efficient I ² C sensor with wide dynamic range (1–65535 lx) and low noise	Reliable for indoor illumination tracking; minor spectral bias mitigated via calibration using TES-1332 A lux meter
DSM501A	Dust (PM _{2.5})	Optical scattering principle affected by high humidity and coarse particle size; reduced accuracy at low concentrations (< 5 $\mu\text{g}/\text{m}^3$)	Widely adopted low-cost particulate sensor; suitable for comparative trend monitoring	Calibration with TSI DustTrak II improved response linearity; residual RMSE (5.1 $\mu\text{g}/\text{m}^3$) acceptable for indoor air-quality evaluation
ESP32-WROOM(core module)	System Controller	Wi-Fi current spikes (up to 500 mA) may cause transient voltage dips if unregulated	High-performance dual-core processor with integrated Wi-Fi/BLE and low power modes	Voltage stabilization via regulated power supply eliminated signal jitter, ensuring consistent sensor operation

methodology described in Sect. 4.7, showed an average node-level transmission latency of 215 ± 40 ms and an end-to-end latency of 310 ± 55 ms. This remarkably fast for communication between nodes and the cloud service, playing a significant role in real-time environmental monitoring. The Packet-level performance was equally robust. A packet delivery ratio of 99.2% was recorded over 48 continuous hours, while a slightly lower value (98.7%) was observed during calibration under higher interference. These harmonized values confirm high reliability and minimal packet loss under real-world operational conditions. Hence, it can be clearly verified that this system performs exceptionally well in terms of low network delay, high transfer reliability, and low energy consumption, which meet the mandatory QoS parameters for efficient IoT-based environmental surveillance.

Although the selected sensors provided satisfactory performance for real-time indoor environmental monitoring, several inherent limitations must be acknowledged for their inclusion in the system (See Table 12). The DSM501A dust sensor, for example, exhibits moderate sensitivity to airflow turbulence and requires a stable particulate concentration for accurate readings. Its infrared scattering principle is also affected by humidity drift, leading to slightly elevated noise levels at higher RH values. Despite these drawbacks, it was selected due to its low cost, wide availability, and adequate precision after calibration, making it appropriate for continuous monitoring in resource-constrained settings.

Table 13 Comparative benchmarking

Feature/Metric	Proposed MCIMS	Awair Omni	Netatmo Smart Indoor Monitor
Measured Parameters	Temperature, Humidity, Light, PM _{2.5}	Temperature, Humidity, CO ₂ , VOC, PM _{2.5}	Temperature, Humidity, CO ₂
Sampling Interval	Configurable (5–60 s)	5 s	~5 min
Communication	Wi-Fi (MQTT)	Wi-Fi (App API)	Wi-Fi
Cloud Access	Open API (ThingSpeak)	Proprietary Dashboard + API	Proprietary App + API
Cost (USD)	45	≈450	≈200
Data Accessibility	Full raw data access	Partial/filtered	Partial
Intended Use	Low-cost research & deployment	Enterprise/clinical IAQ	Consumer IAQ
Calibration Transparency	Full, user-controlled	Black-box	Black-box
Power Usage	198 mW	380 mW	410 mW

The DHT22 and BH1750 sensors also exhibit minor response-time delays and temperature-dependent nonlinearities; however, these were mitigated through preprocessing and calibration. Overall, while each sensor introduces certain limitations, their selection was driven by the need to balance accuracy, low power consumption, cost efficiency, and compatibility with the ESP32-WROOM platform. The impact of these limitations on data reliability is minimal within the scope of indoor trend monitoring, but they should be considered in future applications requiring regulatory-grade measurements.

6.5 Benchmarking against conventional systems

The benchmarking presented in Table 13 is based exclusively on technical specifications obtained from manufacturer datasheets of Awair Omni Model A-2 and Netatmo Healthy Home Coach NHC-EC, rather than physical parallel experimentation. The developed IoT-based indoor monitoring system has provided functionality similar to that of high-end commercial devices, while offering significant advantages in cost efficiency, energy consumption, and data update rate. Wi-Fi connectivity through MQTT enables seamless real-time data transmission and interoperability with open-source cloud platforms, such as ThingSpeak, out of the box, which is in contrast to the proprietary ecosystems used by Awair Omni and Netatmo. Four environmental parameters—temperature, humidity, dust concentration, and light intensity—have been monitored, offering broader coverage of indoor comfort and air quality indicators compared to commercial units that omit either particulate or illuminance sensing. Hence, it enhances the temporal resolution for the tracking of fluctuations in the environment. Besides this, the system has proved to be very energy-efficient and consumed only 198 mW, which is nearly half compared to that by Awair and less than half compared to Netatmo, thus allowing extended battery operation. Economically, it provided a transformative advantage of having a total cost of approximately USD 45, which is nearly one-tenth of the commercial equivalents without any performance compromise.

The baseline comparison demonstrates that the MCIMS platform provides competitive performance for temperature, humidity, illuminance, and PM_{2.5} when benchmarked against Awair Omni and Netatmo reference readings. While the commercial units offer additional CO₂ and VOC capabilities, the MCIMS system benefits from higher sampling granularity, open-access data, and significantly reduced cost. Sensor agreement analysis (RMSE, MAE, R², and confidence intervals) shows that the MCIMS readings fall within

acceptable ranges for low-cost environmental monitoring, thereby validating its viability for research and academic applications.

7 Conclusion

This study presents a validated, low-cost, multi-sensor IoT architecture capable of monitoring key indoor environmental parameters including temperature, humidity, particulate matter, and illuminance with high accuracy and low latency. Through a calibrated sensing pipeline, hybrid cloud-edge communication, and rigorous evaluation of power and network performance, the MCIMS demonstrates its suitability for continuous indoor monitoring in academic and resource-limited environments.

Beyond confirming the technical feasibility of an ESP32-WROOM based IEQ system, this work contributes a reproducible methodological framework that integrates calibration, power modelling, and communication benchmarking elements frequently absent from comparable low-cost monitoring studies. The results collectively highlight the system's potential for scalable deployment in smart-building applications and for supporting real-time environmental analytics.

Future iterations will incorporate additional air-quality indicators such as CO₂, VOCs, and radon, and will include extended deployments across diverse building typologies. A controlled, side-by-side comparison with commercial indoor monitors will further strengthen the validation of the platform's measurement accuracy. These enhancements will enable the MCIMS to evolve toward a fully integrated, intelligent indoor-environment assessment and control system.

7.1 Limitations of current experimentation

The present implementation of the MCIMS was primarily validated in controlled university environments including a postgraduate laboratory, an engineering classroom, and a central library. These settings were deliberately chosen to ensure consistent environmental conditions for sensor calibration, communication benchmarking, and power profiling.

This current system does not incorporate measurements of additional critical indoor air-quality indicators such as carbon dioxide (CO₂), volatile organic compounds (VOCs), and radon gas. These parameters play an important role in assessing indoor ventilation quality and health risk, particularly in enclosed academic or residential environments. Their absence constrains the breadth of environmental insight provided by the system. Future iterations of MCIMS will integrate NDIR-based CO₂ sensors, MOS-based VOC modules, and long-term radon detectors to expand the system's coverage.

However, it is acknowledged that this restricted deployment duration (three days) and controlled occupancy patterns limit the generalizability of results across more dynamic indoor environments such as commercial buildings or residential settings. Factors such as variable ventilation rates, particulate load, and ambient noise could affect long-term sensor stability and communication performance.

Additionally, the current implementation relies entirely on Wi-Fi communication, which may not be optimal in large-scale or industrial deployments where signal congestion, energy consumption, or infrastructure constraints become significant. The absence of low-power wide-area network (LPWAN) testing such as LoRaWAN, NB-IoT, or Sigfox limits insight into long-range, low-power applicability. Future versions will

incorporate hybrid communication support to evaluate performance under different network regimes.

Finally, while the system includes an initial calibration stage, the study does not quantify long-term calibration drift for sensors such as the DSM501A and DHT22. These sensors can experience accuracy degradation due to dust accumulation, humidity exposure, and electronic aging. Periodic calibration checks, drift-compensation models, and automatic recalibration routines will be integrated into future deployments to ensure sustained accuracy.

To address this limitation, the next phase of the research will include extended field deployment across diverse building typologies, with data collection spanning several weeks to capture diurnal and seasonal variations. Additional comparative analysis will be performed against industrial-grade IAQ monitoring systems to further validate measurement consistency and robustness.

7.2 Recommendations and future works

The proposed system can be improved or extended to include more sensors to calculate additional parameters related to the indoors that can help to have a proper understanding of the situation to take correct actions for improving the comfort and health of users.

- **Integration of Machine Learning:** The incorporation of machine learning technology within the existing framework to analyze the acquired data and determine future trends can facilitate the production of analytical results for the control of indoor environments.
- **Sensor Network Enhancement:** A possible area of concern for future researchers could be the improvement of the sensor network to consider parameters like the amount of carbon dioxide, the quantity of radon presents in the room, and air currents. These parameters have a large effect on comfort levels.
- **Implementation of Automated Control Systems:** Designing such automated control systems that respond to data conveyed by sensors to modify HVAC systems, lighting systems, or air purifiers can improve the functionality of the proposed system. Automated systems ensure efficient usage of energy and high standards of environment control.

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Author contributions

MSB: Writing - original draft, Conceptualization; writing, methodology, analysis. AA: Writing - original draft, Supervision, validation, investigation.

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Data availability

The datasets used and/or analyses during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and Consent to participate

This study did not involve human participants, animal subjects, or sensitive biological materials. All procedures conducted in this research were limited to environmental monitoring using electronic sensors in standard indoor spaces. Therefore, ethical approval was not required according to the institutional and national research guidelines. No human participants were directly involved in the study, and no personal or identifiable information was collected. As such, consent to participate is not applicable.

Consent for publication

The authors affirm that the manuscript contains no individual or identifiable personal data. All authors have reviewed the final version of the manuscript and freely give their consent for publication.

Financial interests

The authors declare they have no financial interests.

Competing interests

The authors declare no competing interests.

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